

Exercises 09/08

1. Define $\mathbb{R}^n = \mathbb{R} \oplus \mathbb{R}^{n-1}$ for $n > 2$. Show $\mathbb{R}^n$ is a vector space under coordinate-wise operations.
2. Does $\mathbb{R}^2 \oplus \mathbb{R} = \mathbb{R} \oplus \mathbb{R}^2$?
3. Consider the set $\mathbf{W}$ of vectors in $\mathbb{R}^2$ whose components sum to 1, i.e., $\mathbf{w} \in \mathbf{W}$, where $\mathbf{w} = (w_1, w_2)$ such that $w_1 + w_2 = 1$. Is $\mathbf{W}$ a vector subspace?
4. Is $\{(1, 0, 0), (0, 1, 0), (1, 1, 0)\}$ linearly independent?
5. Show that any set containing $\mathbf{0}$ is linearly dependent.
6. Let $B = \langle \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3 \rangle$ and $\mathbf{v} = (1, 1, 0)$
 - a. Represent $\mathbf{v}$ in terms of B .
 - b. Verify the Exchange Theorem.
7. Is $f(x) = x^2$, a linear map?
8. Show that if $f : \mathbf{V} \rightarrow \mathbf{W}$ is a linear map then $f(\mathbf{0}) = \mathbf{0}$.
9. Let $f : \mathbf{V} \rightarrow \mathbf{W}$ be a linear map. Show that $\ker(f)$ is a subspace of $\mathbf{V}$ and $f(\mathbf{V})$ is a subspace of $\mathbf{W}$.
10. Suppose $\mathbf{V} = \mathbb{R} \oplus \mathbb{R}$ and $\mathbf{W} = \mathbb{R}$. Define $f : \mathbf{V} \rightarrow \mathbf{W}$ by $(v_1, v_2) \rightarrow v_1 + 2v_2$. Find a basis for the kernel of f .
11. Show that the dimension of $\mathbb{R}^n$ is n .